

SQL WINDOWS FUNCTION

1. Name Sub Marks

DT	E	90
LT	M	70
TJ	H	60
SK	E	50

a) $\Rightarrow$ 370

Subject wise total marks

$\rightarrow$ select sub, sum(Marks) from table1

b)

Sub	total marks
E	140
M	70
H	60

$\rightarrow$ select sum(Marks) from table1
 $\swarrow$
 Agg. function

$\rightarrow$ In Windows Function, calculation is performed row by row

a)

Name	Sub	Marks	total marks
DT	E	90	370
LT	M	70	370
TJ	H	60	370
SK	E	50	370

Select Marks | total marks from table

b)

Name	Sub	Marks	total marks
DT	E	90	140
LT	M	70	70
TJ	H	60	60
SK	E	50	140

①
②
③

Advantages of window function

- ① Row by Row calculation
- ② Control the data/window size
- ③ perform row wise comparisons

Syntax of window function

① sum(column) over () $\Rightarrow$ window size is entire table
 $\swarrow$
 function
 • summation

② sum(column) over (partition by column) $\Rightarrow$ window size is grouped / divided / partitioned according to a column
 $\swarrow$ $\swarrow$
 Marks Subject
 • summation

③ sum(column) over (partition by column order by column)

④ sum(column) over (partition by column order by column Range)

Name	Sub	Marks	
DT	E	70	①
TS	E	60	
SK	E	30	
LT	M	70	②
KJ	M	50	
KJ	H	50	③
TS	H	60	

Rows one preceding

a)

Name	Sub	Marks	total marks
DT	E	90	370
LT	M	70	370
TS	H	60	370
SK	E	50	370

→ Select *, Sum(marks) over () as total marks
from table 1

b)

Name	Sub	Marks	total marks
DT	E	90	140
LT	M	70	70
TS	H	60	60
SK	E	50	140

→ Select *, Sum(marks) over (partition by subject)
as total marks from table 1

Types of window function

- ① Agg. window function → Sum, min, max, Avg, Count
↳ Running total
↳ Running average
- ② Ranking window function → Rank, dense Rank, Row number.
- ③ Analytical function → lead, lag, First, last, ntile

Name	Sub	Marks	Running marks
DT	E	90	90
LT	M	70	160
TS	H	60	320
SK	E	50	370

⇓
Sum(marks) over
(Order by Marks)
dense

⇒ Step 1 → DT E 90 90 ✓

Step 2 → DT E 90 90 ✓
LT M 70 160

Step 3 → DT E 90 90 ✓
LT M 70 160
TS H 60 320

Step 4 →

Name	Sub	Marks	
DT	E	90	1
LT	M	70	2
TJ	H	60	3
SK	E	50	4

→ 2nd → LT ⇒

⇒ Rank() over (order by marks)^{desc}